

QMWS Fall 2026: Introduction to Python for Data Analysis

Day Three: Applications

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Section 1

`matplotlib.pyplot`

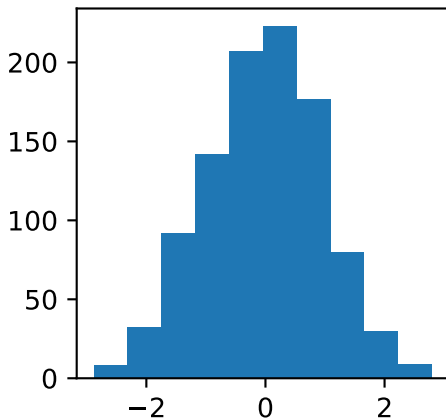
- **matplotlib.pyplot** is a package for visualizing data in Python
- Fun fact: If you have asked ChatGPT for a graph, you likely received one that was made using this package

Basic Histogram

```
import numpy as np
import matplotlib.pyplot as plt

x = np.random.normal(0, 1, 1000)
plt.hist(x)
plt.show()
```

Basic Histogram

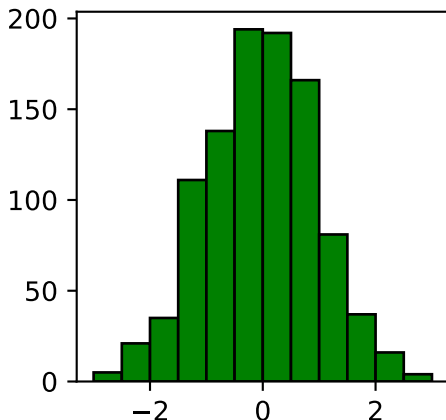


- A few ways to modify the histogram are illustrated below

Modifying the Plot

```
plt.hist(  
    x,                                # Data  
    color = "green",                 # Bar Colour  
    edgecolor = "black",             # Outline Colour  
    bins = np.arange(-3, 3.5, 0.5) # Bins of 0.5 from -3 to +3  
)  
plt.show()
```

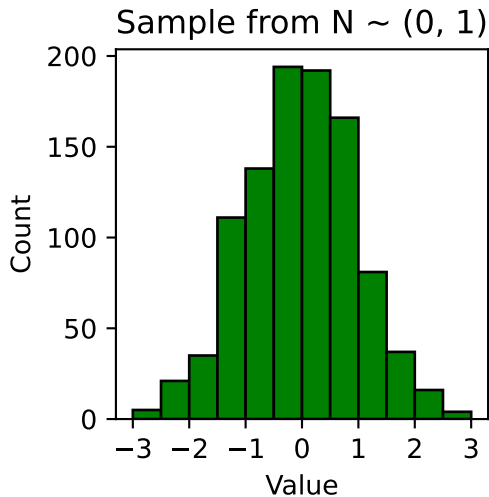
Modifying the Plot



More Ways to Modify

```
plt.hist(  
    x, color = "green", edgecolor = "black",  
    bins = np.arange(-3, 3.5, 0.5)  
)  
  
plt.xticks(np.arange(-3, 4, 1)) # x-axis ticks spaced by 1  
plt.title("Sample from  $N \sim (0, 1)$ ")  
plt.xlabel("Value")  
plt.ylabel("Count")  
plt.show()
```

More Ways to Modify



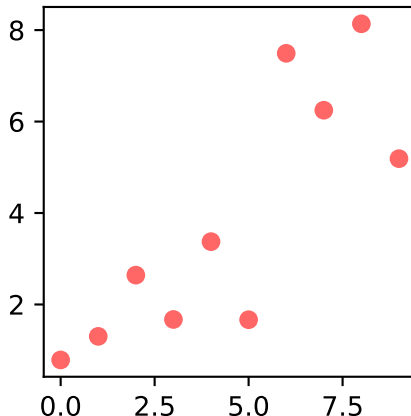
- The below is how data was generated for subsequent scatterplot examples

Simulated Data for a Simple Regression

```
errors = np.random.normal(0, 2, 10) # Normal errors
nums_x = np.arange(0, 10, 1) # Nums 0 to 9
nums_y = nums_x + errors #  $y = x + \text{errors}$ 
```

Scatterplot

```
# plot(x, y, style, transparency)
plt.plot(nums_x, nums_y, 'or', alpha=0.6)
plt.show()
```



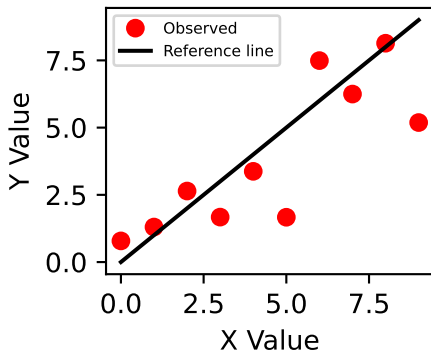
- You can add another plot layer by calling another plotting function before `plt.show()`
- This is useful for adding:
 - reference lines
 - fitted lines
 - multiple groups

Adding a Line to a Scatterplot

```
#Adds observed data as red circles
plt.plot(nums_x, nums_y, 'or')
#Adds reference line (x=y) as black solid line
plt.plot(nums_x, nums_x, '-k')
plt.show()
```

Adding Labels and a Legend

Scatterplot with Reference Line



- `label` gives each layer a name
- `plt.legend()` displays the labels on the plot

Adding Labels and a Legend, continued

- `plt.title()`, `plt.xlabel()`, and `plt.ylabel()` make the plot easier to interpret

Section 2

statsmodels

- **statsmodels** is a package for statistical testing and modelling
- It is useful when you want:
 - formal hypothesis tests
 - regression models
 - p -values and confidence intervals
 - detailed statistical output

Importing statsmodels Functions

```
import statsmodels.formula.api as smf
from statsmodels.stats.weightstats import ttest_ind
```

- **statsmodels.formula.api** contains functions that use formula notation
- It is commonly renamed `smf`
- `ttest_ind` is a function for independent-samples t -tests

Import Structure

```
import statsmodels.formula.api as smf
#      package.module          alias

from statsmodels.stats.weightstats import ttest_ind
#      package.module.submodule      function
```

Statistical Testing

- Statistical tests help us use a **sample** to make inferences about a larger population
- A test usually starts with a **null hypothesis**
 - Often “no difference” or “no relationship”
- We compare the observed data to what would be expected if the null hypothesis were true

Note

- The code runs the test, but the researcher chooses whether the test makes sense

General Workflow

- 1 State the research question
- 2 Identify the level of measurements and study design
- 3 Summarize and visualize the data
- 4 Run the appropriate test
- 5 Interpret the estimate, confidence interval, and p -value

Example Data

- We will use simulated scores from two independent groups
- Each row represents one participant
- The group labels are stored in a pandas DataFrame

Creating the Data

```
import pandas as pd

np.random.seed(42)

df = pd.DataFrame({
    "group": ["control"] * 30 + ["treatment"] * 30,
    "score": np.concatenate([
        np.random.normal(70, 10, 30),
        np.random.normal(76, 10, 30)
    ])
})
```

Inspecting the Data

```
print(df.head())    # First five rows
print(df.shape)     # (number of rows, number of columns)
```

	group	score
0	control	74.967142
1	control	68.617357
2	control	76.476885
3	control	85.230299
4	control	67.658466

(60, 2)

- Always inspect the data before running a test

Independent-Samples t -test

- Used to compare the means of **two independent groups**
- Do not use this test if the same participants appear in both groups
- Two versions exist depending on whether the groups' variances are assumed equal — both use `ttest_ind()`, just with a different `usevar` argument
- We'll work through this test using our general workflow

Research Question:

- Do treatment and control groups have different mean scores?

Null hypothesis:

$$H_0 : \mu_{\text{treatment}} - \mu_{\text{control}} = 0$$

Independent-Samples t -test: Level of Measurement

- group: categorical (nominal), two independent levels — grouping variable
 - Study design: independent (between-subjects) groups
- score: continuous (interval/ratio) — outcome variable

Independent-Samples t -test: Inspecting the Data

```
print(df.groupby("group")["score"].describe().round(2))
```

	count	mean	std	min	25%	50%	75%	max
group								
control	30.0	68.12	9.00	50.87	64.09	67.66	73.60	85.79
treatment	30.0	74.79	9.31	56.40	68.91	75.35	81.45	94.52

- Always re-check the data immediately before running a test
- `groupby("group")` summarizes the control and treatment rows separately
- `.describe()` gives common summary statistics for each group

Selecting Each Group

```
control = df.loc[
    df["group"] == "control", # Rows to include
    "score"                  # Column to return
]
```

```
treatment = df.loc[
    df["group"] == "treatment",
    "score"
]
```

- `ttest_ind()` needs the two groups as separate variables

Independent-Samples t -test: The usevar Argument

```
ttest_ind(  
    x1,                # First group's values  
    x2,                # Second group's values  
    alternative = "two-sided",  
    usevar = "pooled", # or "unequal" (Welch's)  
    value = 0          # Null-hypothesis difference  
)
```

- usevar = "pooled" assumes equal population variances (the regular independent-samples t -test)
- usevar = "unequal" does **not** assume equal population variances (**Welch's t -test**) — a useful default when in doubt

Independent-Samples t -test: The usevar Argument

- Use `alternative = "two-sided"` when either group could have the larger mean
- Use `value = 0` when the null hypothesis is no mean difference

Independent-Samples t -test: Running the Test

```
from statsmodels.stats.weightstats import ttest_ind

pool_result = ttest_ind(
    treatment, control, usevar = "pooled"
)

welch_result = ttest_ind(
    treatment, control, usevar = "unequal"
)
```

- Only the usevar argument changes between the two tests

Independent-Samples t -test: Results

- `ttest_ind()` returns three values: the t -statistic, the p -value, and the degrees of freedom.

```
print(pool_result)
print(welch_result)
```

```
(np.float64(2.82107476851771), np.float64(0.006542565964576836), 58.0)
(np.float64(2.82107476851771), np.float64(0.006544738502602257), 57.93320998252982)
```

You can also index these results:

```
print(f"{pool_result[0]}, "
      f"{pool_result[1]}, {pool_result[2]}")
print(f"{welch_result[0]}, "
      f"{welch_result[1]}, {welch_result[2]}")
```

```
2.82107476851771,0.006542565964576836, 58.0
```

```
2.82107476851771,0.006544738502602257, 57.93320998252982
```

t -Test Interpretation

- The t -statistic describes the group difference relative to its uncertainty
- The p -value is the probability, assuming the null hypothesis is true, of observing a test statistic at least as extreme as the one observed
- The degrees of freedom help determine the t -distribution used by the test

! Important

- A p -value is **not** the probability that the null hypothesis is true

Calculating the Mean Difference

```
mean_difference = treatment.mean() - control.mean()

print(f"Treatment mean: {treatment.mean():.2f}")
print(f"Control mean: {control.mean():.2f}")
print(f"Mean difference: {mean_difference:.2f}")
```

Treatment mean: 74.79

Control mean: 68.12

Mean difference: 6.67

- The t -test output should be interpreted alongside the observed group difference
- The mean difference itself does not depend on the variance assumption, so it is the same for both versions

Interpreting the Results: Equal Variances

```
print(  
    f"Mean difference = {mean_difference:.2f}\n"  
    f"t({pool_result[2]:.2f}) = {pool_result[0]:.2f}\n"  
    f"p = {pool_result[1]:.3f}"  
)
```

Mean difference = 6.67

t(58.00) = 2.82

p = 0.007

- Assumes the two groups have equal population variances

Interpreting the Results: Unequal Variances (Welch's)

```
print(  
    f"Mean difference = {mean_difference:.2f}\n"  
    f"t({welch_result[2]:.2f}) = {welch_result[0]:.2f}\n"  
    f"p = {welch_result[1]:.3f}"  
)
```

Mean difference = 6.67

t(57.93) = 2.82

p = 0.007

- Does **not** assume equal population variances
- Compare the two reports above — here the conclusion is the same either way, which is common when group variances are similar

Correlation

- A correlation describes the direction and strength of a **linear relationship**
- Pearson's correlation coefficient is represented by r
- It ranges from -1 to 1

Research Question:

- Is there a linear relationship between study hours and score?

! Important

- Correlation does not establish causation

Correlation: Level of Measurement

- `study_hours`: continuous (interval/ratio)
- `score`: continuous (interval/ratio)
- Study design: observational/correlational (no groups being compared)
- Both variables are continuous, so a Pearson correlation is appropriate

Adding a Second Variable

```
np.random.seed(7)

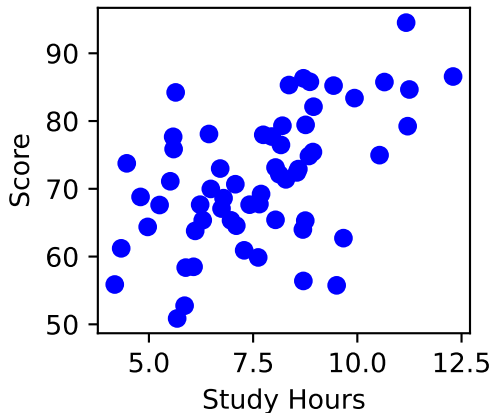
df["study_hours"] = (
    2
    + 0.08 * df["score"]
    + np.random.normal(0, 1.5, len(df))
)
```

Scatterplot

```
plt.plot(df["study_hours"], df["score"], "ob")  
plt.xlabel("Study Hours")  
plt.ylabel("Score")  
plt.show()
```

- Use a scatterplot before calculating a correlation — this is the *inspecting the data* step of our workflow

Scatterplot



.corr() Method

```
correlation = df["study_hours"].corr(  
    df["score"]  
)
```

```
print(f"r = {correlation:.2f}")
```

```
r = 0.53
```

- .corr() describes the observed sample relationship

Correlation: Running the Test

- `.corr()` gives the **estimate** (r) of the linear relationship, but not a p -value or confidence interval on its own
- In **simple linear regression** with one predictor, the slope's significance test is mathematically equivalent to testing whether r differs from zero
- We'll get the confidence interval and p -value for this same relationship next, using regression

Simple Linear Regression

- Simple linear regression examines the relationship between:
 - one continuous **outcome**
 - one continuous **predictor**
- Here, the outcome is `score` and the predictor is `study_hours`

Research Question (same as above):

- Is there a linear relationship between study hours and score?

$$\text{score} = \text{intercept} + (\text{slope} \times \text{study hours}) + \text{error}$$

Simple Linear Regression: Level of Measurement

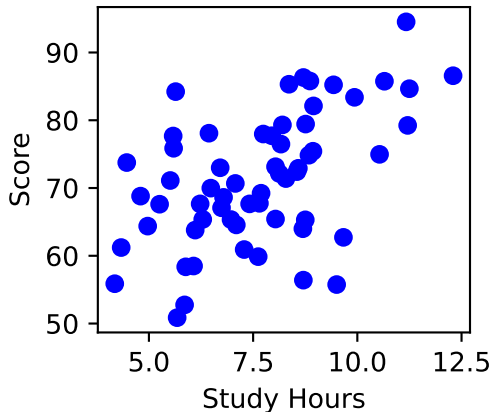
- `study_hours`: continuous (interval/ratio) — predictor
- `score`: continuous (interval/ratio) — outcome
- Study design: observational/correlational, same as above

Simple Linear Regression: Inspecting the Data

```
plt.plot(df["study_hours"], df["score"], "ob")  
plt.xlabel("Study Hours")  
plt.ylabel("Score")  
plt.show()
```

- Same scatterplot as before — always inspect the data before fitting a model

Simple Linear Regression: Inspecting the Data



Formula Notation: Formula Structure

```
"score ~ study_hours"  
# outcome ~ predictor
```

- The outcome goes on the left of ~
- The predictor goes on the right of ~
- Column names must match the DataFrame

Creating a Model with `smf.ols()`

```
model_definition = smf.ols(  
    formula = "score ~ study_hours",  
    data = df  
)
```

```
print(type(model_definition))
```

```
<class 'statsmodels.regression.linear_model.OLS'>
```

- `ols` stands for **ordinary least squares**
- `smf.ols()` defines the model but does not estimate its coefficients yet

Fitting a Model with `.fit()`

```
model = model_definition.fit()
```

```
print(type(model))
```

- `.fit()` estimates the intercept, slope, and other model statistics
- Store the fitted result so its attributes can be inspected

Method Chaining: Defining and Fitting in One Step

```
model = smf.ols(  
    "score ~ study_hours",  
    data = df  
)
```

- This is a common way to define and fit a model

Model Parameters with .params

```
print("Intercept:", round(model.params["Intercept"], 2))  
print("study_hours:", round(model.params["study_hours"], 2))
```

Intercept: 50.43

study_hours: 2.75

- Intercept: predicted score when study hours equals zero
- study_hours: predicted score change for one additional study hour

Accessing One Parameter

```
intercept = model.params["Intercept"]  
slope = model.params["study_hours"]
```

```
print(f"Intercept: {intercept:.2f}")  
print(f"Slope: {slope:.2f}")
```

Intercept: 50.43

Slope: 2.75

- Use the parameter name to extract a specific estimate

Testing the Slope with .pvalues

```
slope_p_value = model.pvalues["study_hours"]  
  
print(f"study_hours p-value: {slope_p_value:.3f}")  
  
study_hours p-value: 0.000
```

Null hypothesis:

$$H_0 : \text{population slope} = 0$$

Confidence Intervals with `.conf.int()`

```
slope_ci = model.conf_int().loc["study_hours"]
```

```
print(f"Lower: {slope_ci[0]:.2f}")
```

```
print(f"Upper: {slope_ci[1]:.2f}")
```

Lower: 1.59

Upper: 3.90

- A confidence interval gives a range of values compatible with the model and data

Model Overview: `.summary().tables[0]`

```
summary = model.summary()
print(summary.tables[0])
```

OLS Regression Results

```
=====
Dep. Variable:          score    R-squared:                0.280
Model:                  OLS      Adj. R-squared:           0.268
Method:                 Least Squares    F-statistic:             22.57
Date:                   Wed, 09 Sep 2026    Prob (F-statistic):      1.37e-05
Time:                   15:14:30    Log-Likelihood:         -210.99
No. Observations:        60    AIC:                    426.0
Df Residuals:            58    BIC:                    430.2
Df Model:                 1
Covariance Type:         nonrobust
=====
```

- Table 0 summarizes the model setup and overall fit
- Useful for checking the outcome, sample size, R^2 , and model type

Model Overview: `.summary().tables[1]`

```
print(summary.tables[1])
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	50.4278	4.553	11.076	0.000	41.314	59.541
study_hours	2.7466	0.578	4.751	0.000	1.589	3.904

- Table 1 gives the estimates for each model term
- Focus first on the coefficient, p -value, and confidence interval columns

Model Overview: `.summary().tables[2]`

```
print(summary.tables[2])
```

```
=====
Omnibus:                0.940    Durbin-Watson:                1.864
Prob(Omnibus):          0.625    Jarque-Bera (JB):          0.843
Skew:                   -0.281    Prob(JB):                  0.656
Kurtosis:               2.852    Cond. No.                  34.0
=====
```

- Table 2 contains diagnostic and assumption-related information
- Useful later when checking whether the model is behaving reasonably

.rsquared

- R^2 is the proportion of outcome variance described by the fitted model
- It ranges from 0 to 1
- A larger R^2 does not prove that the model is correct or causal

```
print(f"R^2: {model.rsquared:.3f}")
```

```
R^2: 0.280
```

Predicted Outcomes with `.predict()`

```
df["predicted_score"] = model.predict(df)

for i in range(3):
    print(
        f"Observed: {df['score'][i]:.2f}; "
        f"Predicted: {df['predicted_score'][i]:.2f}"
    )
```

Observed: 74.97; Predicted: 79.36

Observed: 68.62; Predicted: 69.08

Observed: 76.48; Predicted: 72.86

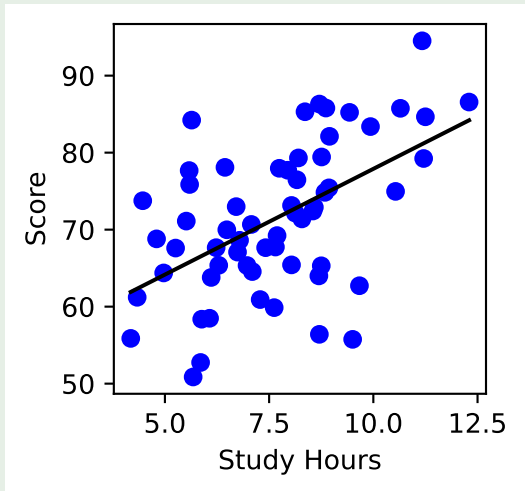
- `.predict()` calculates the fitted outcome for each row

Plotting Model Predictions: Observed and Predicted Values

```
ordered = df.sort_values("study_hours")

plt.plot(df["study_hours"], df["score"], "ob")
plt.plot(
    ordered["study_hours"], ordered["predicted_score"], "-k"
)
plt.xlabel("Study Hours")
plt.ylabel("Score")
plt.show()
```

Fitted Line



- Blue circles are observed values
- The black line contains model-predicted values

Reporting Regression Results: Formatting the Result

```
slope_ci = model.conf_int().loc["study_hours"]

print(f"b = {model.params['study_hours']:.2f}, "
      f"95% CI [{slope_ci[0]:.2f}, {slope_ci[1]:.2f}], "
      f"p = {model.pvalues['study_hours']:.3f}")
```

b = 2.75, 95% CI [1.59, 3.90], p = 0.000

- Report the estimated slope, confidence interval, and *p*-value together
- Use **associated with**, rather than **caused**
- This matches the significance test for the Pearson correlation computed earlier — the slope's *p*-value is mathematically identical to the correlation's *p*-value

Multiple Regression

- Multiple regression extends simple regression to **more than one predictor**
- Predictors can be continuous or categorical — categorical predictors are automatically dummy-coded in formula notation
- Here we add `group` as a second predictor alongside `study_hours`

Research Question:

- Does study hours predict score, after accounting for group — and does the effect of study hours depend on group?

$$\text{score} = \text{intercept} + b_1(\text{study hours}) + b_2(\text{group}) + \text{error}$$

Multiple Regression: Level of Measurement

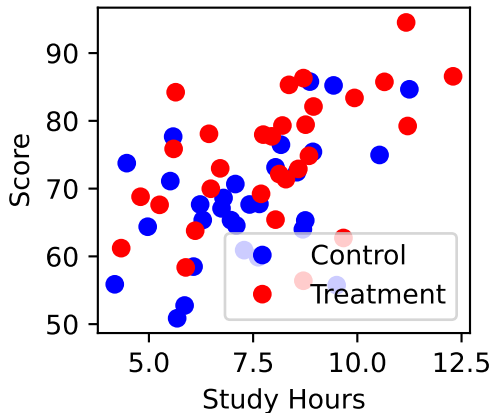
- `study_hours`: continuous (interval/ratio) — predictor
- `group`: categorical (nominal), two levels — predictor (dummy-coded)
- `score`: continuous (interval/ratio) — outcome

Multiple Regression: Inspecting the Data

```
control_rows = df[df["group"] == "control"]
treatment_rows = df[df["group"] == "treatment"]

plt.plot(
    control_rows["study_hours"], control_rows["score"],
    "bo", label = "Control")
plt.plot(
    treatment_rows["study_hours"], treatment_rows["score"],
    "ro", label = "Treatment")
plt.xlabel("Study Hours")
plt.ylabel("Score")
plt.legend()
plt.show()
```

Multiple Regression: Inspecting the Data



Multiple Regression: Formula Notation

```
"score ~ study_hours + group"
```

```
# Main effects only
```

```
"score ~ study_hours * group"
```

```
# Main effects AND their interaction; equivalent to:
```

```
"score ~ study_hours + group + study_hours:group"
```

- + includes each predictor's **main effect** only, holding the other predictor constant
- * includes the main effects **and** the interaction between them

Multiple Regression: Running the Test

```
main_effects_model = smf.ols(  
    "score ~ study_hours + group", data = df  
) .fit()
```

```
interaction_model = smf.ols(  
    "score ~ study_hours * group", data = df  
) .fit()
```

- Only the formula string changes between the two models

Multiple Regression: Interpreting the Results (Main Effects)

```
main_effects_model.summary()
```

```
Intercept: b = 49.68, 95% CI [40.91, 58.45], p = 0.000  
group[T.treatment]: b = 5.09, 95% CI [0.91, 9.26], p = 0.018  
study_hours: b = 2.51, 95% CI [1.38, 3.64], p = 0.000
```

Multiple Regression: Interpreting the Results: (Main Effects)

- Each coefficient is interpreted *holding the other predictor constant*

Multiple Regression: Interpreting the Results (Interaction Effects)

```
interaction_model.summary()
```

```
Intercept: b = 49.73, 95% CI [36.69, 62.77], p = 0.000  
group[T.treatment]: b = 4.99, 95% CI [-  
13.00, 22.98], p = 0.581  
study_hours: b = 2.51, 95% CI [0.77, 4.24], p = 0.005  
study_hours:group[T.treatment]: b = 0.01, 95% CI [-  
2.29, 2.31], p = 0.991
```

Multiple Regression: Interpreting the Results (Interaction Effects)

- Switching + to * added exactly one new term:
`study_hours:group[T.treatment]`
- It tests whether the `study_hours` slope differs between `control` and `treatment`
- Here it is far from significant and barely shifts the other coefficients — evidence the effect of `study_hours` does **not** depend on `group`

Comparing the Two Models: Running the Test

```
from statsmodels.stats.anova import anova_lm
```

```
comparison_table = anova_lm(  
    main_effects_model, interaction_model  
)
```

```
print(comparison_table)
```

	df_resid	ssr	df_diff	ss_diff	F	Pr(>F)
0	57.0	3604.470329	0.0	NaN	NaN	NaN
1	56.0	3604.462629	1.0	0.0077	0.00012	0.991312

Comparing the Two Models: Interpreting the Results

- Tests whether adding the interaction explains significantly more variance in score than the main-effects-only model
- Use this p -value, not the individual interaction term's p -value, to judge whether the interaction belongs in the model
- Here, p is far from significant — the interaction does not improve the model